

GROUP NO. 4

Blazing Trends: A Statistical Exploration of Excessive Heat on the Community

STATISTICS PROJECT 2024-25

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PURPOSE/RELEVANCE OF THE SURVEY:

- Studying excessive heat in Mumbai is crucial due to the city's rapid urbanization and dense population, which exacerbate heat stress on infrastructure and public health. The purpose of this statistical analysis is to investigate the effects of excessive heat on various aspects of human lifestyle and health. By examining data on temperature extremes, health outcomes, and behavioral changes during increasing heat periods, this study aims to identify significant correlations and trends. The ultimate goal is to understand how excessive heat influences public health, daily activities and thereby providing insights that can inform public health policies, emergency response strategies, and community preparedness plans.

TARGET POPULATION AND SAMPLE SIZE:

- Mumbai city population of age bracket 15 years and above - 60 years
At Least 250 responses (Standardized)
- Sample size: 30

OBJECTIVES OF THE SURVEY:

- To study the relation between lifestyle and excessive heat
- To determine how the excessive heat has affected people's health.
- To assess the level of public awareness regarding the impacts of excessive heat.

METHODOLOGY:

- Sampling technique used : Non probability- Convenience and Snowball Sampling.
- We did stratify our population with each section of our questionnaire.
- Developed a structured questionnaire with a multiple choice, scale based, multiple select and open ended questions.
- Conducted pre-testing by administering the questionnaire to a small group of the residents of Mumbai to identify any issues with questioning, comprehension, or formatting. The changes identified with the help of the pre-testing questionnaire were made and the final questionnaire was created.
- For our study we have mentioned that the questions in the Google form has to filled according to the experience they had during the months of March to june 2024.
- We administered our questionnaire by sending out a Google form and shared the survey link through email and social media platforms like Instagram, LinkedIn and Whatsapp.
- The questions included were about demographics (age, gender, employment status and region of residence).
- We received a total of 287 responses (including 30 respondents of pilot survey)
- We performed multiple statistical tests like Correlation, Correlation Matrix (Python), Mantel–Haenszel, Odds Ratio, F-Exact Test (R Programming), Z-Test, Assignment Problem

PILOT SURVEY REPORT:

Significance of conducting the pilot survey:

To improve the quality of the questionnaire by identifying any ambiguities or logistical issues

To assess the data collection process and the response rates

Sample size: 30

Data collection method : Online google form

Duration of data collection : 1 week

Changes made after analysis of pilot survey :-

Removing email id requirement since some respondents wanted their information to be confidential.

In question 11 an additional option of “Not Applicable” was added since there are some households that don’t make use of an AC.

In question 12 an additional option of “ I don’t use private transport”, was added for people who might not use any private transportation.

In question 20, option “others” converted to descriptive to know what other precautionary measures subjects take .

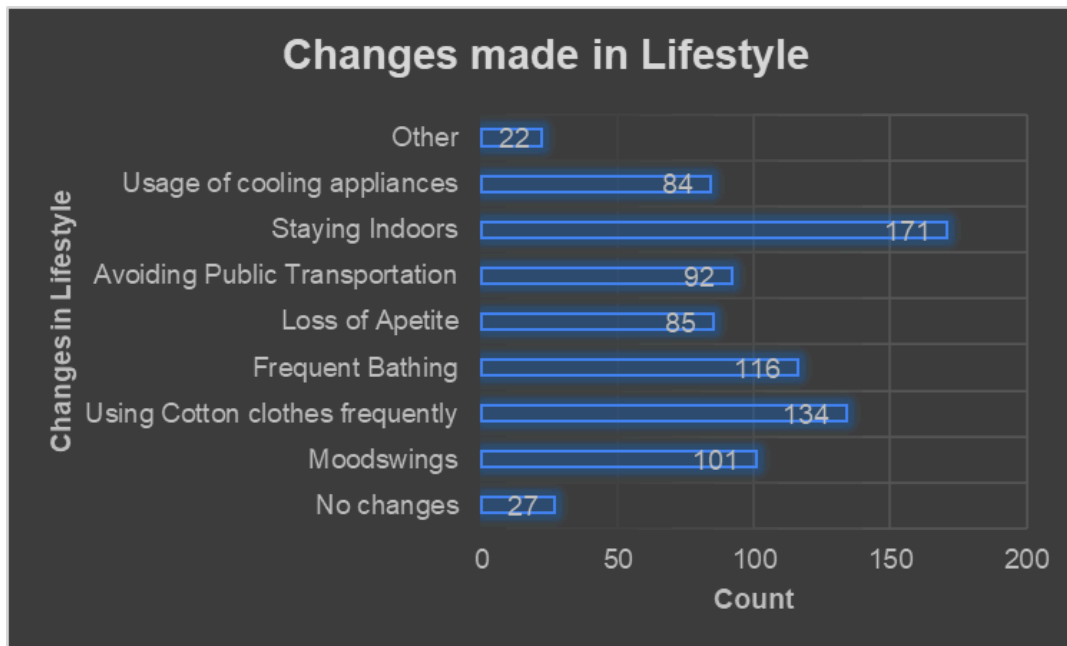
Changing the option type of question 21 from multiple choice to multiple select as subjects might suggest more than 1 way in which they want to see the changes incorporated by local authorities .

By incorporating these valuable recommendations, the main questionnaire was made more comprehensive, ensuring a thorough understanding of participants' perceptions, concerns, and insights on the impact of excessive heat on community

ANALYSIS:

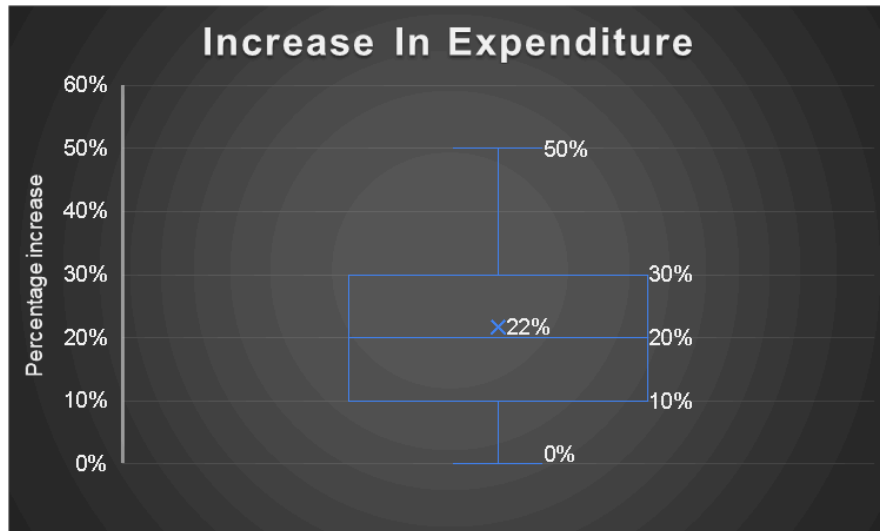
OBJECTIVE 1: To study the relation between lifestyle and excessive heat

→ CHANGES MADE TO LIFESTYLE:



- BOX AND WHISKER PLOT FOR INCREASE IN EXPENDITURE

The increase in expenditure due to excessive heat responses were 0%,10%,20%,30%,40% and 50%.



Analysis of the box and whisker plot for expenditure increase :

- The typical increase in expenditure (median) is 22%.
- On average, expenditure increased by 21%.
- Most (50%) of the data falls within a range of 10% to 30% increase in expenditure. This range is called the interquartile range (IQR).
- There are no extreme changes in expenditure since there are no outliers identified in the data set.
- In simpler terms, half of the data points show an increase between 10% and 30%, with a typical increase around 22%. The average increase is slightly higher at 23%, but there are no very high or low spending changes compared to the rest of the data.

- **Correlation Matrix between Employment status and increase in expenditure due to excessive heat**

Employment status - Independent Variable

Increase in Expenditure - Dependent variable

Increase in Expenditure in percentage						
Employment Status		10	20	30	40	50
	student	12	33	15	6	5
	unemployed	2	4	2	2	2
	working (Indoors)	23	30	24	10	7
	working (Outdoors)	10	30	19	6	8

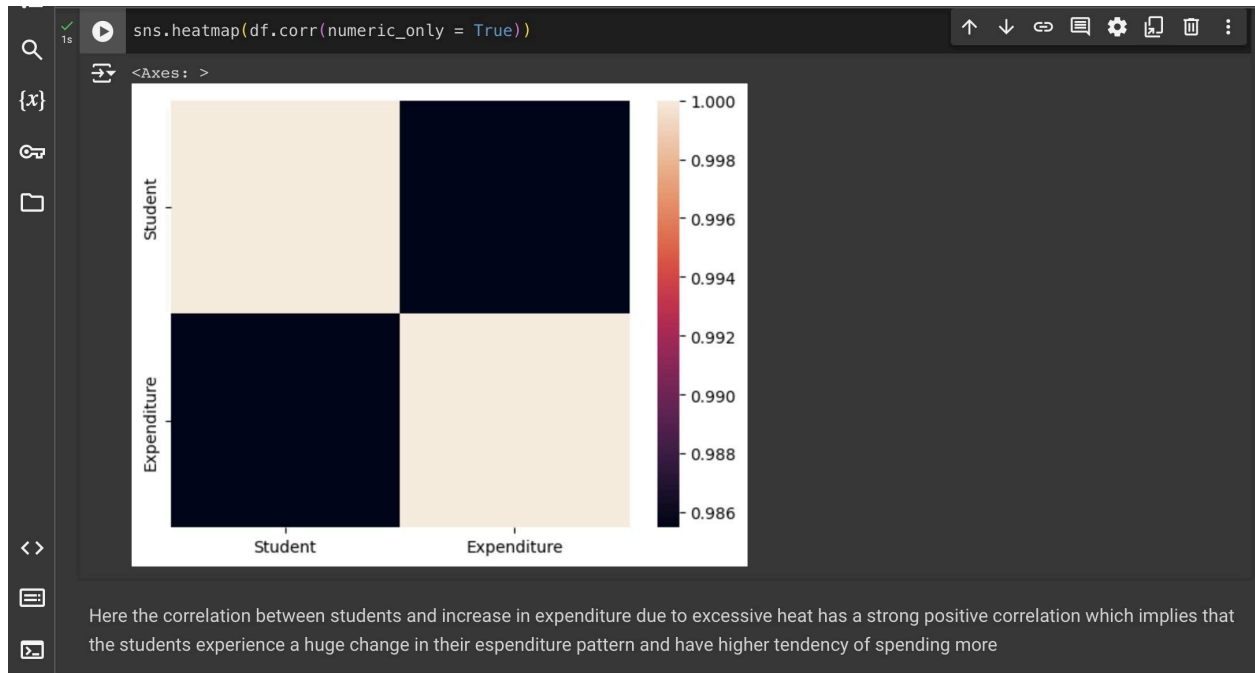
```
data = {"Student": [12,33,15,6,5],
        "Expenditure": [47,97,60,24,22]}
df = pd.DataFrame(data,columns=['Student','Expenditure'])
df
```

	Student	Expenditure
0	12	47
1	33	97
2	15	60
3	6	24
4	5	22

Next steps: [Generate code with df](#) [View recommended plots](#) [New interactive sheet](#)

```
[3] corrMatrix = df.corr()
print(corrMatrix)
```

	Student	Expenditure
Student	1.000000	0.985478
Expenditure	0.985478	1.000000



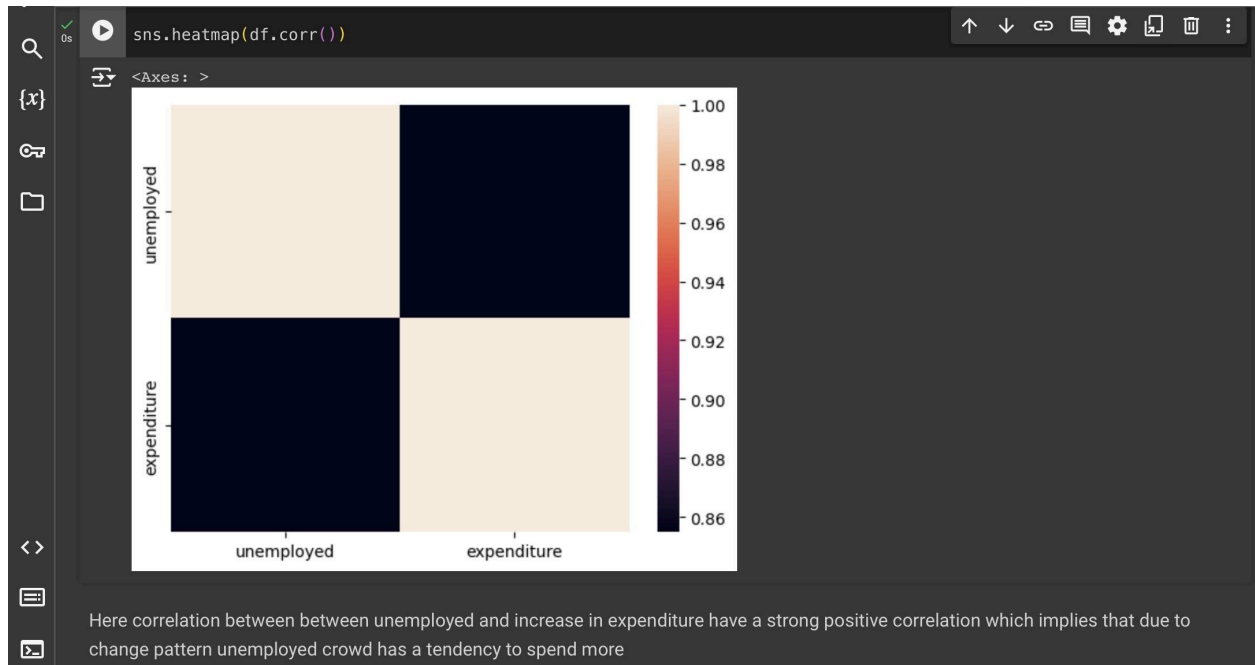
0s `data1={'unemployed':[2,4,2,2,2],
'expenditure':[47,97,60,24,22]}
df=pd.DataFrame(data1,columns=['unemployed','expenditure'])
df`

	unemployed	expenditure
0	2	47
1	4	97
2	2	60
3	2	24
4	2	22

Next steps: [Generate code with df](#) [View recommended plots](#) [New interactive sheet](#)

0s `df.corr()`

	unemployed	expenditure
unemployed	1.000000	0.854913
expenditure	0.854913	1.000000



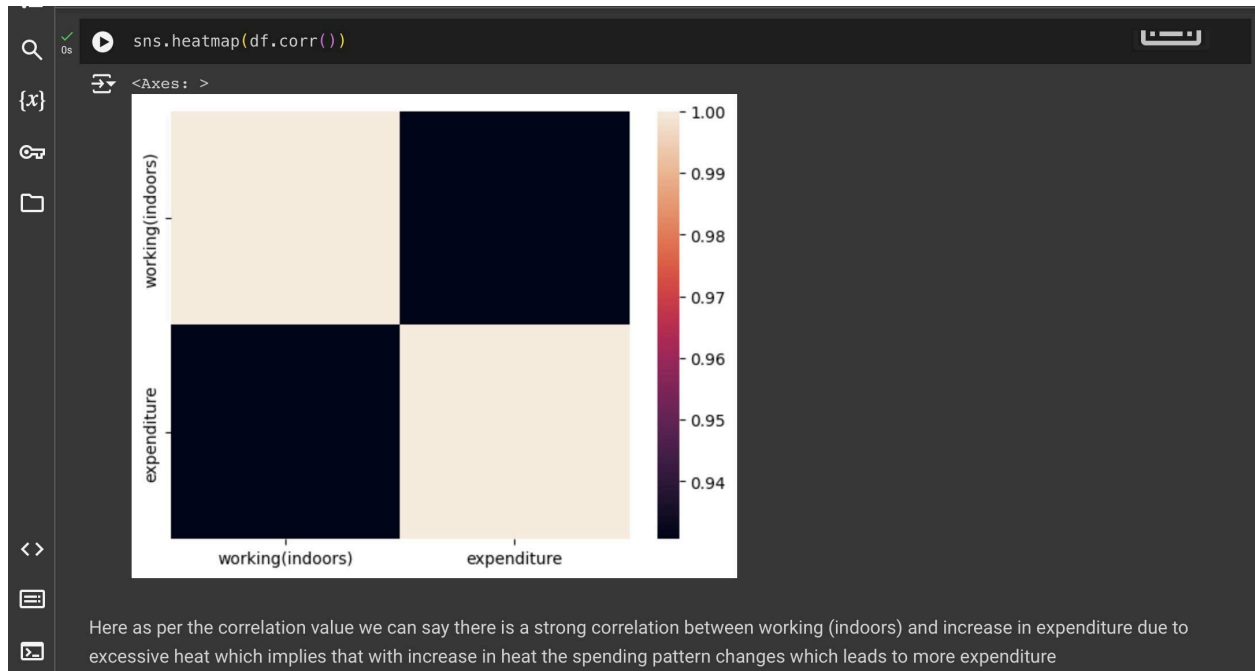
0s `data2={'working(indoors)': [23,30,24,10,7],
'expenditure': [47,97,60,24,22]}
df=pd.DataFrame(data2,columns=['working(indoors)','expenditure'])
df`

	working(indoors)	expenditure
0	23	47
1	30	97
2	24	60
3	10	24
4	7	22

Next steps: [Generate code with df](#) [View recommended plots](#) [New interactive sheet](#)

0s `df.corr()`

	working(indoors)	expenditure
working(indoors)	1.000000	0.930633
expenditure	0.930633	1.000000



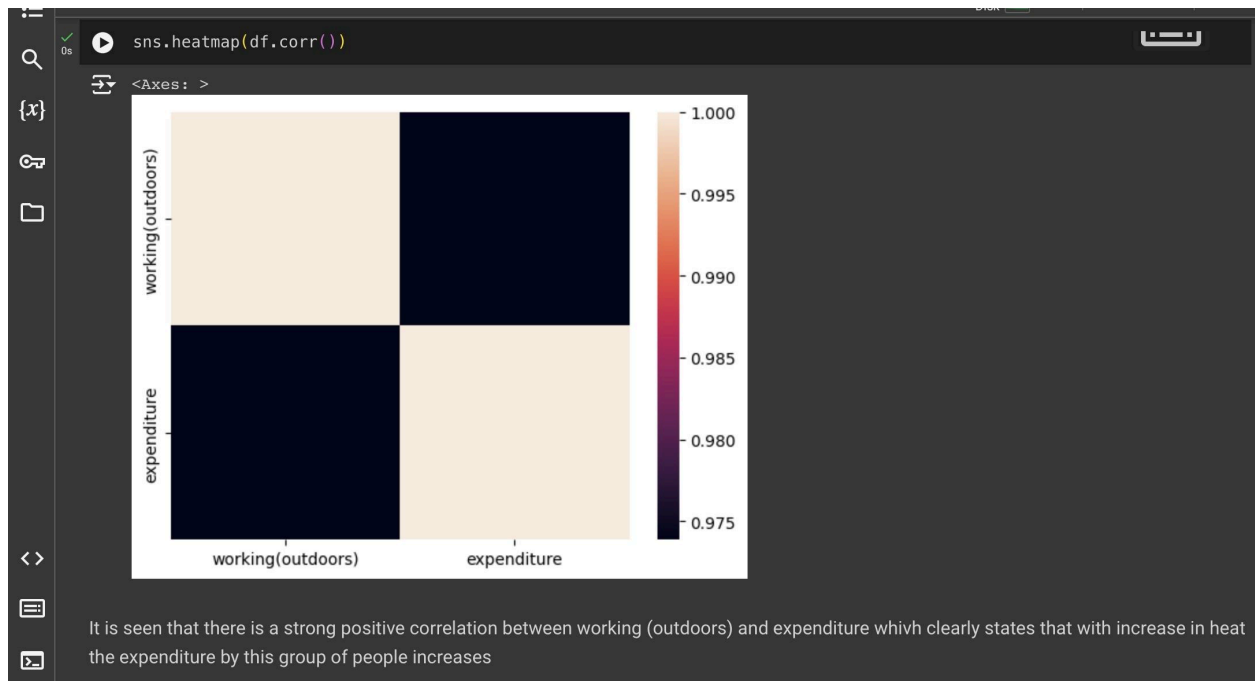
0s [11] `data3={'working(outdoors)': [10,30,19,6,8],`
`'expenditure': [47,97,60,24,22]}`
`df=pd.DataFrame(data3,columns=['working(outdoors)','expenditure'])`
`df`

	working(outdoors)	expenditure
0	10	47
1	30	97
2	19	60
3	6	24
4	8	22

Next steps: [Generate code with df](#) [View recommended plots](#) [New interactive sheet](#)

0s `df.corr()`

	working(outdoors)	expenditure
working(outdoors)	1.000000	0.973885
expenditure	0.973885	1.000000



CONCLUSION:

Conclusion: As seen above for people from different employment status their relation with increase in expenditure is strong positive correlation which implies increase in one leads to increase in other that is with increase in excessive heat the expenditure will automatically increase for each and every one irrespective of their employment status in different extent there may be difference in extent of increase in expenditure but there is going to be an increase in expenditure with increase in excessive heat.

SECONDARY DATA REPORT:

Extreme heat waves, exacerbated by climate change, significantly impact expenditure patterns. Households face higher energy bills due to increased cooling demands, while businesses incur additional costs for mitigating heat-related productivity losses. Consumer preferences shift towards cooling products and heat-resistant items. Moreover, the rise in heat-related illnesses strains healthcare systems, leading to higher medical expenses. These combined factors contribute to substantial economic burdens, emphasizing the urgent need for climate action to mitigate the financial consequences of extreme heat events.

- Assignment Problem of temperature of area and AC consumption hours

PROCEDURE:

Collected the values for the increased number of hours of AC consumed per day by people from different regions where no. Of hours ranging from 1 to 5 and region under consideration being south bombay, Mumbai suburban , Thane/kalyan , NAVI Mumbai/ Panvel , Vasai Verar .

For temperature we have taken the average temperature in the region under study during months of March to June .

Taken temperature as rows and increased No. of hours of Ac consumption per day as column and value as count of people using those many hours of AC for respective temperature (according to region)

First Assignment Problem table:

		<i>No of hours of AC consumption increased per Day</i>				
Region	Average	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>	<i>5</i>

	temperature during march to June for respective regions					
South Bombay	32	2	7	7	7	10
Mumbai suburban	34	2	12	24	19	40
Thane / Kalyan	37	2	9	13	11	27
Vasai / virar	38	1	4	2	3	6
Navi Mumbai / panvel	39	1	4	3	10	6

After solving this Assignment Problem Using Hungarian Method we get:

ASSIGNMENTS:

	<i>Number of Hours of AC consumption increased per hour</i>					
Region	Average temperature during march to June for respective regions	1	2	3	4	5
south Bombay	32	3	0	1	0	2
Mumbai suburban	34	0	0	13	8	27
Thane / Kalyan	37	2	0	4	2	16

Vasai / virar	38	7	1	0	0	1
Navi Mumbai / panvel	39	6	0	0	6	0

Through this AP we can conclude that for the respective temperature the optimum Number of hours of AC consumption is as given below:

Region	Temperature(in degree Celsius)	No of AC hours
south Bombay	32	4
Mumbai suburban	34	1
Thane / Kalyan	37	2
Vasai / virar	38	3
Navi Mumbai / panvel	39	5

By using the above mentioned table as a guide we can optimize AC consumption.

To understand how the AP has helped us achieve a better AC consumption plan we show the difference using carbon emission pattern :

CARBON EMISSION:

Secondary Data : 1 hr of AC consumption releases 0.625 to 1 pound of CO₂ into the atmosphere.

Applying here,

(Observed through primary data)	No. of Hours of AC consumption per day	No. of people
South bombay	2.69	33
Mumbai suburban	3.48	97
Thane / Kalyan	3.57	62

Vasai / virar	3.10	16
Navi Mumbai / panvel	3.41	24

AC CONSUMPTION:

$$W = \frac{\sum_{i=1}^n w_i X_i}{\sum_{i=1}^n w_i}$$

$$= (3.41 \times 24 + 3.48 \times 97 + 2.69 \times 33 + 3.10 \times 16 + 3.57 \times 62) / 232$$

$$= 3.35 \text{ hrs of AC consumption per day}$$

(Revised Data after performing AP)	No. of Hours of AC consumption per day	No. of people
South bombay	4	33
Mumbai suburban	1	97
Thane / Kalyan	2	62
Vasai / virar	3	16
Navi Mumbai / panvel	5	24

AC CONSUMPTION :

$$W = \frac{\sum_{i=1}^n w_i X_i}{\sum_{i=1}^n w_i}$$

$$=(4 \times 33 + 1 \times 97 + 2 \times 62 + 3 \times 16 + 5 \times 24) / 232$$

$$= 2.24 \text{ hrs per day AC consumption}$$

CONCLUSION:

By the observed data we have seen the average number of increased hours of AC consumption is 3.35 hrs per household per day so total amount of carbon emission is $3.34 \times 1 = 3.34$ pounds of carbon dioxide per household

If we see according to the optimum Numbers of increased hours of AC consumption as obtained from Assignment problem we get an average of 2.24 hrs of AC per day per household so the total amount of carbon emitted per household is 2.24 pounds

Clearly,

$$3.35 - 2.24 = \mathbf{1.11 \text{ pounds}}$$

By using the AC for Optimum Number of hours for respective temperature as obtained through Assignment Problem we can reduce a total of 1.11 pounds of carbon emission.

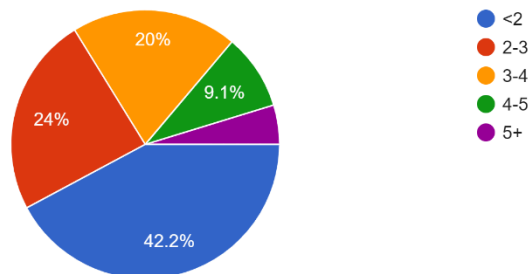
- Regression between Hours of Exposure under the Sun and Increase in usage of Private Transportation

Hours of sun exposure - Independent variable

Increased usage of private transport - Dependent variable

6.No. of hours you are exposed to the sun in a day?

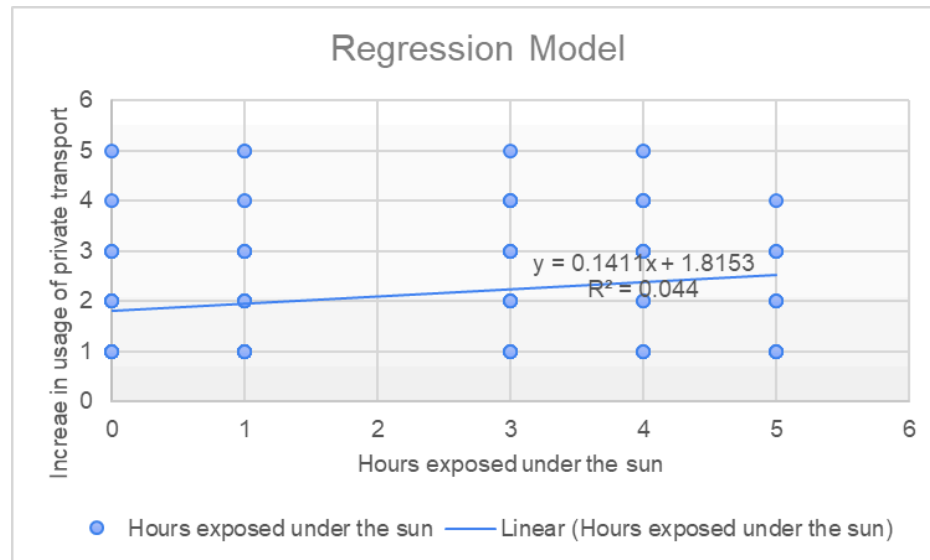
275 responses



INCREASE IN USE OF PRIVATE TRANSPORT

NO of times	Count
don't use private transport	75
5+ times	30
4-5 times	42
3-4 times	58
2>=	82

1	SUMMARY OUTPUT									
2										
3	<i>Regression Statistics</i>									
4	Multiple R	0.209931								
5	R Square	0.044071								
6	Adjusted R	0.040121								
7	Standard Error	1.721006								
8	Observations	244								
9										
10	<i>ANOVA</i>									
11		<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>				
12	Regression	1	33.04522	33.04522	11.15691	0.000969				
13	Residual	242	716.7704	2.961861						
14	Total	243	749.8156							
15										
16		<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>	
17	Intercept	1.414124	0.225071	6.283021	1.53E-09	0.970777	1.857472	0.970777	1.857472	
18	X Variable 1	0.311194	0.093166	3.340196	0.000969	0.127673	0.494714	0.127673	0.494714	
19										
20										



Conclusion:

- The regression model has a weak explanatory power ($R^2 = 0.0441$), meaning that the model only explains about 4.41% of the variance in private transport usage based on hours exposed to the sun.
- The model as a whole is statistically significant (F-statistic and p-value).
- The coefficient for hours exposed to the sun is positive and statistically significant, suggesting that more hours of sun exposure are associated with an increase in private transport usage. However, the overall model fit is quite low, indicating that **other factors may be influencing private transport usage that are not included in this model.**

- Spearman's Rank Correlation Coefficient between Sleep and Productivity

The question was asked how the excessive heat has impacted sleep, and the options given were sleep increased, decreased and no impact, and they were ranked accordingly. The question about impact on productivity was scale based, so we clubbed 1 and 2 scale as low productivity, 3 as medium productivity and 4 and 5 as high productivity. The ranks were given according to the counts of individuals who chose low, medium and high productivity.

Sleep	Increased	Decreased	No Impact
(X)	129	46	68

Productivity	Low	Medium	High
(Y)	98	74	71
Rank of X	1	3	2
Rank of Y	1	2	3
d (Rank X-Y)	0	1	-1
d^2	0	1	1

$$\rho = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)}$$

$$= 0.5$$

Interpretation:

A Spearman's correlation coefficient 0.5 indicates a moderately positive relationship between sleep duration and productivity. This means that as sleep duration increases, there's a general trend for productivity to also increase, but the relationship is not perfectly linear. People who tend to sleep more tend to also be more productive, but it's not a perfect match. There are some exceptions to this pattern, but overall, getting enough sleep seems to help people get more done.

Conclusion:

The correlation shows that there's a general trend for people who sleep longer to be more productive, but it's not a guaranteed relationship. Some people might be naturally productive even with less sleep, while others might find it difficult to be productive with less sleep.

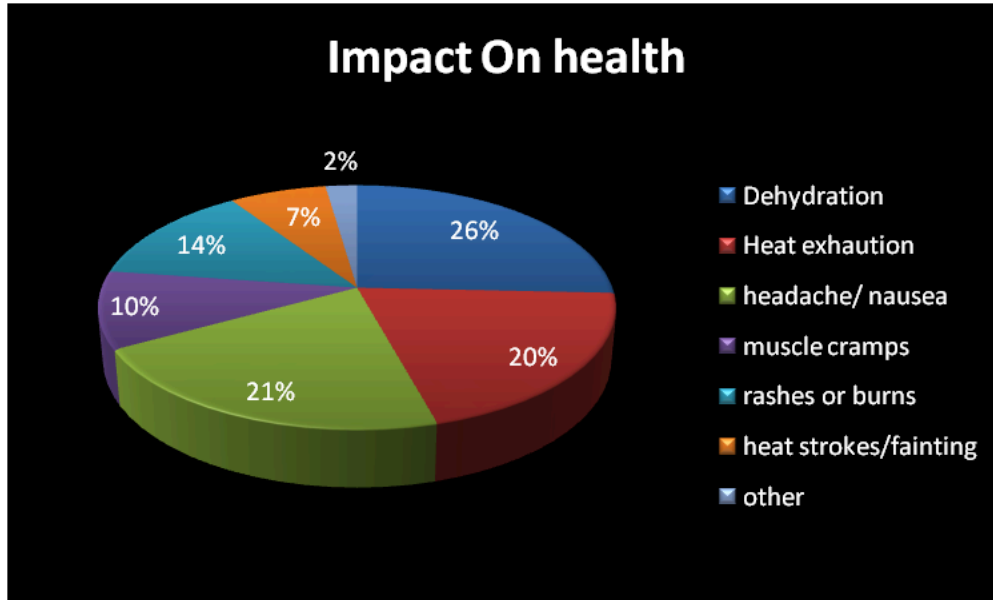
Secondary reference:

- ❖ Most individuals experience a decline in productivity when they don't get enough sleep, but the exact amount of sleep needed to maintain optimal performance can differ based on individual factors like genetics, lifestyle, and overall health; generally, consistent quality sleep is crucial for peak productivity across individuals.

- ❖ Even if someone gets enough sleep, poor quality sleep (frequent waking, disrupted sleep cycles) can still negatively impact productivity.

OBJECTIVE 2: To determine how the excessive heat has affected people's health.

→ IMPACT ON HEALTH:



- **ODDs RATIO** between no. of hours exposed to the sun and the severity of symptoms

Null Hypothesis (H0): There is no association between sun exposure and illness severity.

Alternative Hypothesis (H1): There is an association between sun exposure and illness severity.

The respondents were asked how many hours a day they spend in the sun, and they were given the options <2, 2-3, 3-4, 4-5, and 5+. Thus, we clubbed <2 and 2-3 as low exposure, and 3-4, 4-5, and 5+ as high exposure. Similarly for severity of symptoms (scale based), 1 and 2 scales are considered as mild symptoms, while 3–5 scales are considered as severe symptoms.

SUN EXPOSURE	Severity of Symptoms		TOTAL
	MILD	SEVERE	
LOW	44	57	101
HIGH	18	43	61
TOTAL	62	100	Σ162

The odds of severe illness for the high sun exposure group

$$\begin{aligned} &= \frac{(\text{Number of severe cases with high sun exposure})}{(\text{Number of mild cases with high sun exposure})} \\ &= 43 / 18 \end{aligned}$$

The odds of severe illness for the low sun exposure group

$$\begin{aligned} &= \frac{(\text{Number of severe cases with low sun exposure})}{(\text{Number of mild cases with low sun exposure})} \\ &= 57 / 44 \end{aligned}$$

The odds ratio = $\frac{(\text{Odds for high sun exposure group})}{(\text{Odds for low sun exposure group})}$

$$= 1.208$$

Interpretation:

Odds Ratio (OR) > 1: This indicates that individuals with high sun exposure are **more likely** to develop severe illness compared to those with low sun exposure.

Conclusion:

The data suggests that there's a link between sun exposure and the severity of illness. People who spent more time in the sun were more likely to experience severe symptoms.

Secondary reference:

The relationship between sun exposure and the severity of symptoms is directly proportional: the more sun exposure one receives, the more severe the symptoms

- ❖ Mild sun exposure: May result in mild redness and slight discomfort.
- ❖ Moderate sun exposure: Can cause noticeable redness, pain, and potential swelling.
- ❖ Excessive sun exposure: May lead to severe blistering, significant pain, and potential for long-term skin damage like premature aging and increased risk of skin cancer.

- **ASSIGNMENT PROBLEM: Severity of Symptoms and Recovery measure**

Procedure: Collected the values of severity of symptoms caused due to heat and the recovery methods employed by people to recover from the same where severity of symptoms ranged from 1 to 5 and various recovery methods included staying hydrated, rest and light medication, seeking professional medical help and staying indoors. Severity of symptoms was expressed in rows whereas the various recovery methods in columns and value as count of people incorporating a specific recovery method for given severity level.

FIRST AP TABLE:

Severity of Symptoms	Recovery Measures				
	Hydration	Rest and light Medication	Professional Medical Attention	Staying Indoors	Others (open ended option)
1	10	7	2	7	19
2	32	18	7	32	6
3	48	36	16	48	8
4	23	17	13	17	11
5	1	2	2	1	7

ASSIGNMENTS (After solving using Hungarian method)

Severity of Symptoms	Recovery Measures				
	Hydration	Rest and light Medication	Professional Medical Attention	Staying Indoors	Others (open ended option)
1	3	5	0	0	9
2	20	5	0	20	4
3	32	19	5	32	0
4	7	0	12	7	0
5	0	0	6	0	15

Result:

For people with severity of symptoms 1, staying indoors is the most effective recovery method.

For people with severity of symptoms 2, professional medication proved to be the most effective.

For people with severity of symptoms 3, since others was an open ended option people have suggested different methods other than the ones mentioned.

For people with severity of symptoms 4, rest and light medication proved to be most effective.

For people with severity of symptoms 5, hydration proved to be the most effective.

SECONDARY DATA

- Common symptoms among these conditions include dehydration, nausea/headache, muscle cramps, and heat exhaustion. However, heat stroke is uniquely characterized by a high fever and altered mental state.
- The recommended recovery methods for all three conditions involve staying hydrated, resting, and seeking professional medical help when necessary. Mild cases of heat cramps or exhaustion may be treated with home remedies like staying indoors and taking over-the-counter medication. However, heat stroke requires immediate medical attention .
- In conclusion, while the symptoms and treatment for heat-related illnesses share similarities, the severity of heat stroke necessitates prompt medical intervention. Proper hydration, rest, and seeking medical advice are crucial for recovery from all three conditions.

- **ASSIGNMENT PROBLEM: Number of Symptoms and Recovery Measures**

- **Procedure:** Collected the values of count of symptoms caused due to heat and the recovery methods employed by people to recover from the same where count of

symptoms ranged from 1 to 5 and various recovery methods included staying hydrated, rest and light medication, seeking professional medical help and staying indoors.

Count of symptoms was expressed in rows whereas the various recovery methods in columns and value as count of people incorporating a specific recovery method for given severity level.

Count of Symptoms	Recovery Measures				
	Hydration	Rest and light Medication	Professional Medical Attention	Staying Indoors	Others
1	18	15	5	10	3
2	36	11	5	32	4
3	33	25	12	30	8
4	20	20	10	21	10
5	8	9	5	11	5

ASSIGNMENTS (After solving using Hungarian method)

Count of Symptoms	Recovery Measures				
	Hydration	Rest and light Medication	Professional Medical Attention	Staying Indoors	Others

1	8	26	23	10	0
2	4	0	14	9	0
3	0	0	7	5	2
4	0	22	20	0	3
5	0	0	0	0	2

Result:

For people with count of symptoms 1, since others was an open ended option people have suggested different methods other than the ones mentioned.

For people with count of symptoms 2, rest and light medication proved to be the most effective recovery method.

For people with count of symptoms 3, hydration proved to be the most effective recovery method.

For people with count of symptoms 4, staying indoors proved to be the most effective recovery method.

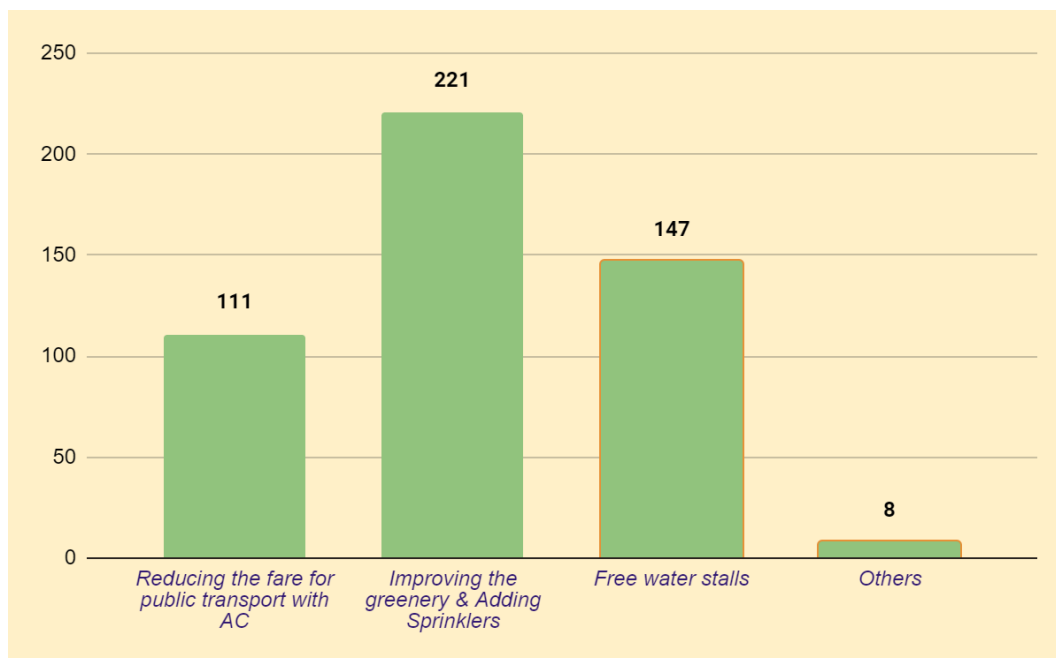
For people with count of symptoms 5, professional medical attention proved to be the most effective recovery method.

SECONDARY DATA

- Studies have consistently linked heat-related illnesses to dehydration, electrolyte imbalance, and elevated body temperature. The recommended recovery methods, such as hydration, rest, and seeking medical attention, are effective in addressing these underlying causes.
- While the provided information is comprehensive, it's important to note that individual experiences may vary. Factors such as age, overall health, and severity of heat exposure can influence the specific symptoms and recovery process. It's always advisable to consult with a healthcare professional for personalized guidance.
- In conclusion, the analysis of the provided data confirms the accuracy of the listed symptoms and recovery methods for heat-related conditions. The findings align with established medical knowledge and practices.

OBJECTIVE 3: To assess the level of public awareness regarding the impacts of excessive heat.

→ Respondents suggestion of change to help cope with increasing heat



- **CROSS TABULATION (Employment status and precautionary measures)**

Employment Status	Precautionary Measures
-------------------	------------------------

	Stayin g Hydrat ed	Using Shield s	Limite d Outdo or expos ure	Keeping supplement ary aid
Retired	9	5	9	4
Student	54	46	30	15
Unemployed	5	5	4	1
Working (Indoors)	62	32	62	11
Working (Outdoors)	58	20	19	38

Results:

- The cross tabulation shows the frequency of different precautionary measures taken by individuals based on their employment status. Staying hydrated and limited outdoor exposure were the most common precautionary measures adopted by the people in the retired group .
- Staying hydrated and using shields were the most common precautionary measures adopted by people belonging to both student and unemployed groups.
- Among people working indoors, staying hydrated and limited exposure to sun (outdoors) were the most common precautionary measures.
- Among people working outdoors staying hydrated and keeping supplementary aid with them were the most precautionary measures adopted.

- **Mantel Haenszel to test the relationship between no of symptoms (risk) and preventative measures (risk factor/exposure) in gender (confounding factor)**

Hypothesis

Null Hypothesis (H0): There is no association between no. of symptoms and preventative measures in gender.

Alternative Hypothesis (H1): Not H0

<3- Less Symptoms (count of symptoms)

Females

Females	Less Symptoms	More Symptoms	Total
Precautions	84	22	106
No precautions	11	4	15
Total	95	26	121

$$n1=121$$

$$e1=(106)(95)/121 = 83.2231405$$

$$v1=(106)(15)(26)(95)/(121^2)(120) = 2.235332286$$

Males

Males	Less Symptoms	More Symptoms	Total
Precautions	93	11	104
No precautions	13	2	15
Total	106	13	119

$$n2=119$$

$$e2=(104)(106)/119 = 92.63865546$$

$$v2= (104)(15)(13)(106)/(119^2)(118) = 1.286464735$$

Chisq Test

Test Statistic

Calc X^2_{MH} $(177-175.861796)^2/$ gives
3.521797021 0.3678543735
 $\sim X^2_{(1)}$

Tab $X^2_{(1)} = 3.841$

Decision Criteria

Calc < Tab Do not
 Reject H_0

ie. There is no association between no of symptoms and taking preventative measures

! BUT WE USE FISHER'S EXACT TEST since $E_i < 5$ (for both tabulations)

Using 2 tailed Hypergeometric Distribution

In R Programming

```
1 #Stats Project
2 #Fishers Exact Test in package tidyverse, summarytools, gtsummary
3 #Males
4 data=matrix(c(84,11,22,4),c(2,2))
5 data
6 fisher.test(data)
7 #Females
8 data1=matrix(c(93,13,11,2),c(2,2))
9 data1
10 fisher.test(data1)
```

Females Fisher's Exact Test for Count Data

data: data

p-value = 0.7368

alternative hypothesis: true odds ratio is not equal to 1

95 percent confidence interval:

0.2930951 5.2782036

sample estimates:

odds ratio

1.3844

Males Fisher's Exact Test for Count Data

data: data1

p-value = 0.668

alternative hypothesis: true odds ratio is not equal to 1

95 percent confidence interval:

0.1261848 7.0433291

sample estimates:

odds ratio

1.2976

In both cases, Do not Reject H_0

since $P \text{ value} > 0.05$

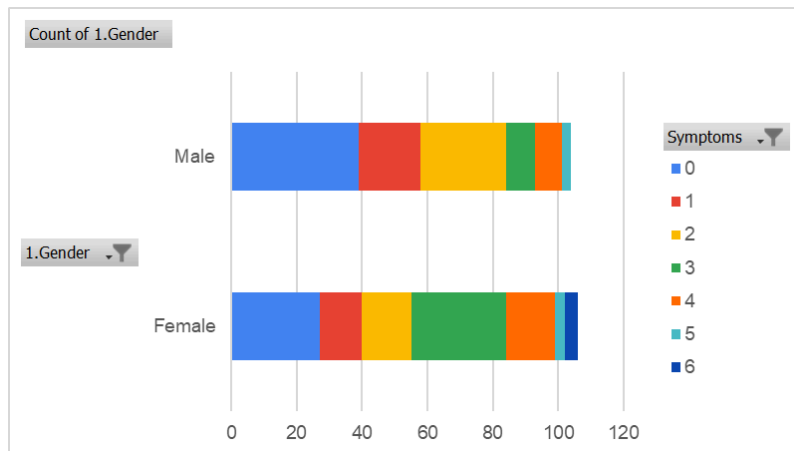
ie. There is no association between no of symptoms and taking preventative measures

This is a limitation of our sample since the majority of our respondents work indoors and hence preventative measures do not play a major/significant role in their symptoms. This could also be because in our sample, the minority that doesn't take preventative measures also show fewer symptoms.

In general however, it is worth noting that a significantly larger number of people of both genders are taking precautions.

Graph :

For Gender (females and males both have taken precautions) and No. of symptoms



More Females took precautions than Males

On average, Females who took precautions still experienced more symptoms than Males who did the same

Corroborate with Secondary data :-

Studies reveal a troubling gender disparity in India when it comes to coping with extreme heat. According to the analysis, women are significantly more vulnerable to extreme temperature conditions than men.

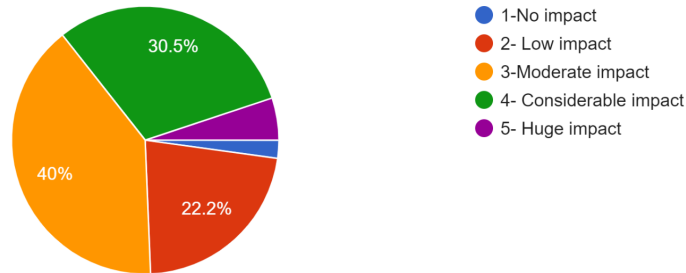
Since 2005, data shows a concerning rise in heat-related deaths among women in India, according to the analysis published in [Significance Magazine](#), a journal by Royal Statistical Society.

“We find through descriptive statistical analysis that publicly available data for India shows that females exhibit a heightened susceptibility to extreme temperature conditions (specifically heat) compared to males,” Ramit Debnath, university assistant professor and an academic director at the University of Cambridge, and one of the authors, told Down To Earth.

□ Age groups and Impact on Wellbeing (Difference in means)

By Central Limit Theorem for a large sample, we use Normal Distribution
T Test → Z-Test (Normal Distribution)

17.How impactful do you think the increasing excessive heat is to your well-being?
275 responses



<3 - low impact

>=3 - High Impact. (On a scale of 1 to 5)

Hypothesis

Null Hypothesis (H0): There is no significant difference in impact means between the two age groups.

Alternative Hypothesis (H1): Not H0

z-Test: Two Sample for Means

	16-35	Age 36+
Mean	3.198275862	3.118110236
Known Variance	0.786432	0.739908
Observations	116	127
Hypothesized Mean Difference	0	
z	0.714012558	
P(Z<=z) one-tail	0.237609707	
z Critical one-tail	1.644853627	
P(Z<=z) two-tail	0.475219414	
z Critical two-tail	1.959963985	

Decision Criteria :

At 95% CI

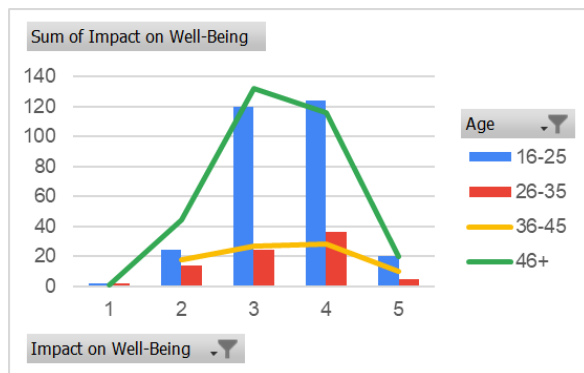
$Z = 0.05$

P-value > Z (both tail types)

Do not Reject H₀

ie there is no sign difference between impact effect for both age groups

ie. And on average, both age groups experience a slight considerable impact (characteristic of a classic bell curve for Z-test, with slight left (negative) skewness corresponding with our objectives and secondary data on excessive heat, as shown below)



Also, in general we can observe (from means) that in age 36+ , people experience negligibly less impact than in age 16-35

But, It is Fact that older generations are more vulnerable to the extremities of excessive heat
This bias may be because although older generations don't FEEL they're symptoms until later/
have delayed responses with age.

This indicates a negligible bias in our sample since we do not have a substantial number of senior citizens/retirees

This could also be because in our sample of older age group, majority work indoors.

Corroborate with Secondary data:-

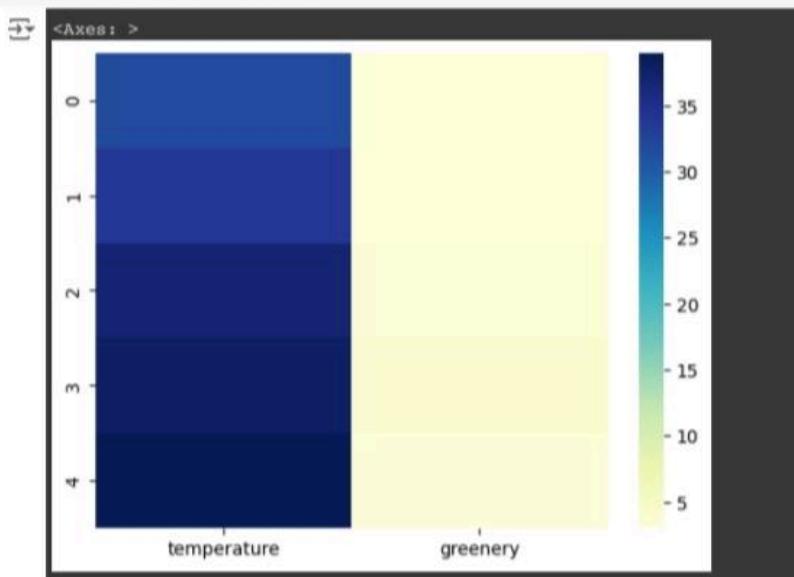
1. Everyone you know will eventually be highly vulnerable to extreme heat

Elaborates that senior citizens do not FEEL their symptoms until much later (due to natural causes), when it may have already come into effect. This leads them to be at a greater physical disadvantage and at a much higher risk.

HEAT MAP :

Region	Average temperature during march to June for respective regions	Greenery on scale(1 to 5)
south Bombay	32	3
Mumbai suburban	34	3.03
Vasai/ virar	37	3.29
NAVI Mumbai/ Panvel	38	4
Thane/kalyan	39	3.2

```
sns.heatmap(df, cmap="YlGnBu")
```



Visual representation of the greenery (scale) for each region in Mumbai, corresponding to their average temperatures, from March-June.

CONCLUSION:

OBJECTIVE 1: To study the relation between lifestyle and excessive heat.

Our analysis revealed that there is a significant impact of excessive heat on lifestyle, with notable increases in energy consumption and transportation usage. Households increased AC usage by an average of 23%, leading to a substantial rise in carbon emissions.

Additionally, private transport usage rose by approximately 4.41% due to increased sun exposure, indicating a shift in travel patterns to avoid the heat. While the relationship between sun exposure and private transport usage is statistically significant, other factors likely influence transportation choices as well. Furthermore, the correlation matrix demonstrates a strong positive relationship between employment status and increased expenditure due to excessive heat, indicating that individuals across various employment sectors experienced rising costs associated with the heatwave. Overall, these findings underscore the substantial impact of excessive heat on daily life and the need for effective strategies to mitigate its effects.

OBJECTIVE2: To determine how the excessive heat has affected people's health.

Based on our analysis, It was seen that there is a correlation between sun exposure and the severity of illness. Individuals who spent more time in the sun were more likely to experience severe symptoms. The calculated odds ratio of 1.208 suggests that individuals with high sun exposure are more likely to develop severe illness compared to those with low sun exposure. Additionally, the analysis of symptom severity and recovery methods revealed that specific strategies, such as hydration, rest and light medication, and seeking professional medical attention, may be more effective for different symptom levels.

These findings highlight the importance of sun protection measures, adequate hydration, and seeking medical attention when necessary to reduce the negative impacts of excessive heat on health.

OBJECTIVE 3 : To assess the level of public awareness regarding the impacts of excessive heat.

Through our analysis we have observed that the majority of respondents, regardless of employment status, took proactive steps to protect themselves from excessive heat, with staying hydrated and limiting outdoor exposure being the most common precautionary measures. While the choice of specific measures varied slightly based on employment status, there was a general awareness of the need to take precautions. However, our analysis revealed no significant association between the adoption of these measures and the severity of heat-related symptoms. This might be due to factors like the predominance of indoor workers in our sample or the relatively low number of respondents who didn't take precautions. Additionally, our statistical analysis suggests that both age groups experienced a similar level of impact from excessive heat, with no significant differences observed. Overall, the findings highlight the importance of public awareness about heat-related risks and the need for continued efforts to promote preventive measures, especially among vulnerable populations.

LIMITATIONS:

In our survey the sample size was 250 . Amongst these 250 responses the distribution of responses from different age groups, employment status, gender and region of residence are not equally represented because of which in many of our analyses the relation between the dependent and the independent variable defer from our secondary data references.

- In the Assignment problem of temperature of different regions and increased number of AC hours per day , Number of symptoms and Recovery methods and severity of symptoms and recovery method we can see that some of the assignments that we have got seems to be a misfit.

As in the Assignment problem of temperature of different regions and increased number of AC hours per day we can see for 32 degrees the optimum increased number of hours is shown to be 4 hours while for 37 degrees it's 2 hours and for 38 degrees it's 3 hours.

And for the Assignment Problem Number of symptoms and Recovery methods we can see for a count of 2 symptoms the most effective way is shown to be rest and light medication and for 3 symptoms is hydration.

For the Assignment Problem of severity of symptoms and recovery method we can see for on a scale of 1 to 5 severity for a severity of 2 the most effective method is shown to be seeking professional medical attention while for 5 it's hydration.

- In our Mantel Haenszel test we get no association between the number of symptoms and precautionary measures (for gender). This goes against the secondary data, that states that those who take precautions must experience lesser symptoms and vice-versa.
- In the Normal test of mean score of well being in younger age group and older age group we can see that the impact on well being for older age group is negligibly less compared to the younger age group.
- This is a limitation of our sample since the majority of our respondents work indoors- preventative measures do not play a significant role in their symptoms. This could also be because in our sample, the minority that doesn't take preventative measures also show fewer symptoms.

- **In Z-test**, It is Fact that older generations are more vulnerable to the extremities of excessive heat (Secondary Data). This indicates a bias- which may be because although older generations don't FEEL their symptoms until later/have a delayed response. (Secondary Research). This also indicates a limitation in our data- did not have a substantial number of senior citizens/retirees and in our sample of older age group, majority work indoors.

REFERENCES :

- World Health Organization (WHO)
- John Hopkins Medicine
- Intergovernmental Panel on Climate Change (IPCC)
- Significance Magazine, a journal by Royal Statistical Society
- Japan Times
- ResMed
- National Library of Medicine

Questionnaire

Demographic:-

1. Age

16-25

26-35

36-45

46+

2. Gender

Male

Female

Other

Prefer not to say

3. Occupation

Student

Working (Indoors)

Working (Outdoors)

Unemployed

Retired

4. Which region of Mumbai do you reside in?

- South Bombay
- Mumbai suburbans
- Thane/Kalyan
- Navi Mumbai/ panvel
- Vasai / virar

Relation between lifestyle and excessive heat

5. Have you felt the need to alter your lifestyle due to excessive heat this year?
Yes/No
6. Which of the following applies to the change in your lifestyle due to excessive heat?
Moodswings
Using Cotton Clothes frequently
Frequent Bathing
Loss of appetite
Avoiding Public Transportation
Staying Indoors more
Bought cooling appliance
Other
7. No. of hours you are exposed to the sun in a day?
<2
2-3
3-4
4-5
5+
8. By how much has your monthly expenditure increased due to excessive heat this year?
(eg. On electricity, appliances, clothing, medicinal products etc)
0
10%
20%
30%
40%
50% +
9. Can you rate the decline in your productivity due to excessive heat?
Scale (1-no decline, 5-most decline)
10. "The excessive heat has impacted my sleep."
Yes Duration of sleep increased, yes duration of sleep decreased, no
11. By how many hours has the AC consumption increased in your household/day?
1
2
3
4

5+

Not Applicable

12. By how much has your usage of private transport increased to avoid the heat? (/week)

2>= times

3-4 times

4-5 times

5+ times

I don't use private transport

Impact on health

13. Has the excessive heat noticeably affected your health? Yes/No.

14. If yes, which of these are applicable?

Dehydration

Heat Exhaustion

Headache/Nausea

Muscle Cramps

Rashes/Burn

Heat Stroke/Fainting

Other

15. How would you rate the severity of your symptoms? (Scale)

1-NA

2- low

3-moderate

4-bad

5-worst

16. How do you typically recover from heat-related illnesses?

Hydration

Rest and light Medication

Professional Medical Attention

Staying Indoors

Other

Precautions & Awareness

17. Are you aware that excessive heat has an impact on your well-being? (Yes/No)

18. How impactful do you think the increasing excessive heat is to your well-being?

1-No impact

- 2- low impact
- 3-moderate impact
- 4- considerable impact
- 5- huge impact

19. Do you take precautionary measures? (Yes/No)

20. If yes, which of these apply?

- Staying hydrated
- Using shields (umbrellas, caps etc)
- Limited outdoor exposure
- Keeping supplementary aid
- Others (descriptive)

21. According to you what measures should the local authorities take to help residents cope with the increasing heat? (Descriptive- Compulsory)

- Reducing the fare for public transport with AC
- Improving the greenery & Adding Sprinklers
- Free water stalls
- Others _____

22. On a scale of 1 to 5, how would you rate the greenery in your residential neighborhood ?

- 1 - very less greenery
- 2 - less greenery
- 3 - moderate greenery
- 4 - good amount of greenery
- 5- very green

Confidentiality Note:

Dear Participants,

Thank you for your participation in our study. We want to assure you that the privacy and confidentiality of your information are of utmost importance to us. Your responses will remain anonymous, as no personally identifiable information will be collected. All collected data will be securely stored and accessible only to authorized researchers involved in the project. Your responses will be used solely

for statistical analysis and academic research, and the findings will be reported in aggregate form to ensure individual responses cannot be identified. Rest assured that your data will not be shared with any third parties, including educational institutions, government agencies, or commercial entities.

Your participation is entirely voluntary, and you have the right to withdraw at any time without any consequences. We appreciate your valuable participation in this study.

Thank You Note

Dear Participants,

We sincerely thank you for your participation in our study. Your valuable input will greatly contribute to our understanding of this topic. Rest assured that your privacy and confidentiality are of utmost importance to us, and all data collected will be handled with strict confidentiality measures. Your involvement is greatly appreciated, and your contributions will significantly impact our research. Thank you for being an essential part of this study.

Acknowledgement -

We would like to express our sincere gratitude to our professors, **Mrs. S.Annapurna** and **Mr. Omkar Thakur** for their invaluable guidance throughout this project. Their expertise in statistics and unwavering support greatly enriched our understanding of the subject matter and encouraged us to explore new concepts.

We are also grateful to our classmates who were always willing to engage in discussions, exchange ideas, and provide feedback. Their collective input helped

us refine our project and consider different perspectives, contributing to its overall quality.

We would like to acknowledge the resources available at the college library and online databases, which provided us with access to a wealth of information and research papers that were crucial for our project.

Additionally, we extend our appreciation to our friends and family for their encouragement and understanding during this demanding period of our studies. Their moral support and patience were instrumental in helping us stay focused and motivated.

Last but not least, we want to acknowledge the dedication and hard work of our team, as this project has been a significant learning experience for us. It has helped us develop valuable skills in data analysis and statistics that we will carry forward into our academic and professional journey.

In conclusion, we are thankful for the support and resources that have contributed to the successful completion of our statistics project. This project has been a valuable experience, and we look forward to applying the knowledge and skills gained from it in our future endeavors.

